

PRANAV JAIN

University of Southern California

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[Website](#) , [Scholar](#)

RESEARCH INTERESTS

My research lies at the intersection of **physics-based simulation**, **scientific machine learning**, **numerical analysis**, and **geometry processing**. I develop **mesh- and grid-free methods** for solving partial differential equations on complex geometric domains. I am particularly interested in the **convergence**, **accuracy**, and **reliability of neural PDE solvers**. I am also exploring **optimal transport**-based formulations for physical simulation.

EDUCATION

University of Southern California (USC) Doctor of Philosophy (PhD), Computer Science	August 2023 - Present <i>Los Angeles, USA</i>
New York University (NYU) Master of Science (MS), Scientific Computing	September 2021 - May 2023 <i>New York, USA</i>
Indraprastha Institute of Information & Technology Delhi (IIITD) Bachelor of Technology with Honors (B-Tech Hons), Computer Science and Applied Mathematics	August 2016 - August 2020 <i>New Delhi, India</i>

PUBLICATIONS

Pranav Jain, Navami Kairanda, Peter Yichen Chen, & Oded Stein. (2026). PINNsur: Physics-Informed Neural Networks for PDEs on Curved Surfaces. [\[link\]](#)

Pranav Jain, Ziyin Qu, Peter Yichen Chen, and Oded Stein. 2024. Neural Monte Carlo Fluid Simulation. In ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24). [\[link\]](#)

Liam Martin, **Pranav Jain**, Zachary Ferguson, Torkan Gholamalizadeh, Faezeh Moshfeghifar, Kenny Erleben, Daniele Panozzo, Steven Abramowitch, Teseo Schneider. A systematic comparison between FEBio and PolyFEM for biomechanical systems. In Computer Methods and Programs in Biomedicine, 2023. [\[link\]](#)

Zachary Ferguson, **Pranav Jain**, Denis Zorin, Teseo Schneider, and Daniele Panozzo. High-Order Incremental Potential Contact for Elastodynamic Simulation on Curved Meshes. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). [\[link\]](#)

Pranav Jain, Munawar Hasan, Donghoon Chang. Spy based analysis of selfish mining attack on multi-stage blockchain. In Cryptology ePrint Archive, 2019. [\[link\]](#)

RESEARCH EXPERIENCE

Google Summer of Code [link] <i>Contributor</i>	May 2026 - August 2026 <i>Remote</i>
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Advisor: Prof. Martin Skrodzki, Dr. Andreas Fabri

- Implemented the algorithm proposed in *Lipschütz et al. (2024). Manifold Meshing of Point Clouds with Guaranteed Smallest Edge Length* for mesh reconstruction from scratch and integrated it into the Computational Geometry Algorithms (CGAL) library

Adobe Research <i>Research Intern</i>	May 2025 - August 2025 <i>Seattle, USA</i>
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Advisor: Dr. Timothy Langlois, Dr. Danny Kaufman, Dr. Jui-hsien Wang

- Researched on how to make generative-AI models more physics accurate by limiting hallucinations for video generation

University of Southern California

PhD Student

August 2023 - Present

Los Angeles, USA

Advisor: Prof. Oded Stein, Prof. Krishna Garikipati

- Researching on empirically proving convergence rates of Poisson problem using Physics Informed Neural Networks (PINN) with respect to the size of the architecture
- Researched and published on a discretization-free fluid simulation method using combination of Monte-Carlo and neural fields

New York University

Research Assistant

September 2021 - May 2023

New York, USA

Advisor: Prof. Daniele Panozzo, Prof. Denis Zorin

- Researched and compared the simulation abilities of PolyFEM and FEBio in terms of accuracy, time and user inputs
- Introduced a high-order finite element formulation for elastodynamic simulation with contact handling

nTopology

Software Engineer Intern in Geometry Team

June 2022 - August 2022

New York, USA

Advisor: Suraj Musuvathy, Ranjeeth Mahankali

- Designed and implemented a method to efficiently stitch different CAD models to form one unified CAD structure

Freie Universität Berlin [\[link\]](#)

Research Intern

September 2020 - August 2021

(Virtual) Berlin, Germany

Advisor: Prof. Konrad Polthier, Dr. Sunil Kumar Yadav

- Researched on a robust and parameter-tuning free method to denoise a point cloud

Fields Undergraduate Summer Research Programme 2020 [\[link\]](#)

Research Intern

July 2020 - August 2020

(Virtual) Toronto, Canada

Advisor: Prof. Thomas Uchida

- Researched on the problem of inverse mechanism synthesis – given a path, design a mechanism that traces the path

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Research Assistant

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Kaushik Kalyanaraman

- Developed DECAT, a software framework for discretizations of the objects and operators of exterior calculus

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Undergraduate Thesis

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Donghoon Chang

- Proved the selfish mining attack strategy on the Bitcoin blockchain for multiple mining pools
- Designed and proved an extended attack for multi-stage blockchain

ACADEMIC COMMUNITY WORK

Reviewing

- SIGGRAPH 2026 February 2026
- SIGGRAPH Asia 2025 July 2025
- Pacific Graphics (PG) 2025 July 2025

Volunteering

- Teaching assistant for the Poisson Reconstruction Project at Summer Geometry Initiative (SGI) 2024 July 2024 - August 2024
- Former member of the Conference Coffee team at SIGGRAPH Research Career Development Committee. 2021-2022

TEACHING EXPERIENCE

University of Southern California

- Teaching Assistant: CSCI 532 - Innovation for Defense Applications Spring 2026
- Teaching Assistant: CSCI 103 - Introduction to Programming Fall 2025
- Teaching Assistant: CSCI 580 - 3D Graphics and Rendering Spring 2025
- Teaching Assistant: CSCI 596 - Scientific Computing and Visualization Fall 2024
- Teaching Assistant: CSCI 104 - Data Structures and Object Oriented Design Spring 2024

New York University

- Grader: MATH 263.3 - Applied Partial Differential Equations Spring 2023
- Grader: MATH 252 - Numerical Analysis Spring 2023
- Grader: MATH 263 - Partial Differential Equations Fall 2021